LECTURE-9

Cellular respiration

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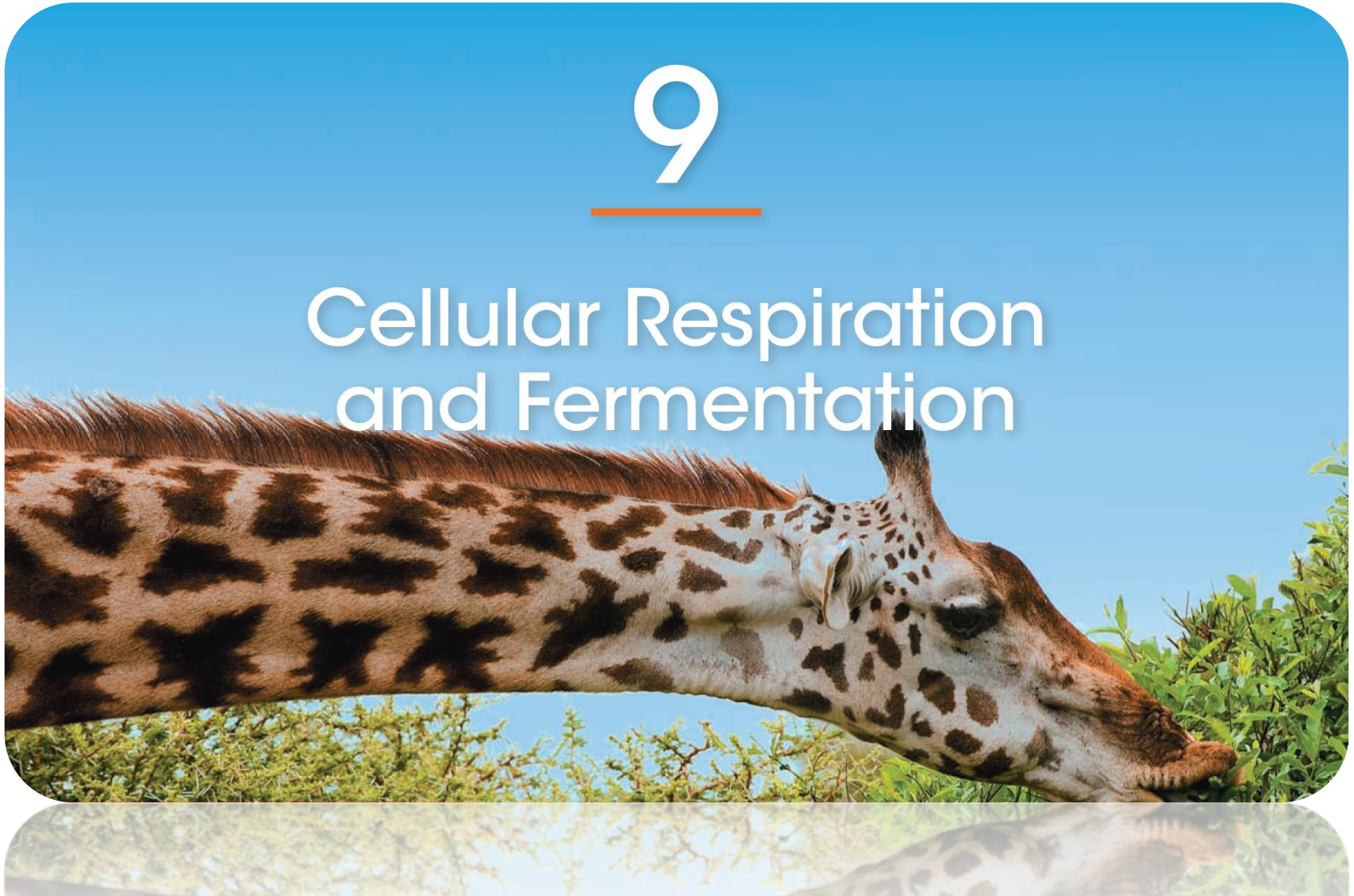
DELHI

Cellular respiration



KEY CONCEPTS

9.1 Catabolic pathways yield energy by oxidizing organic fuels

9.2 Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

9.3 After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules

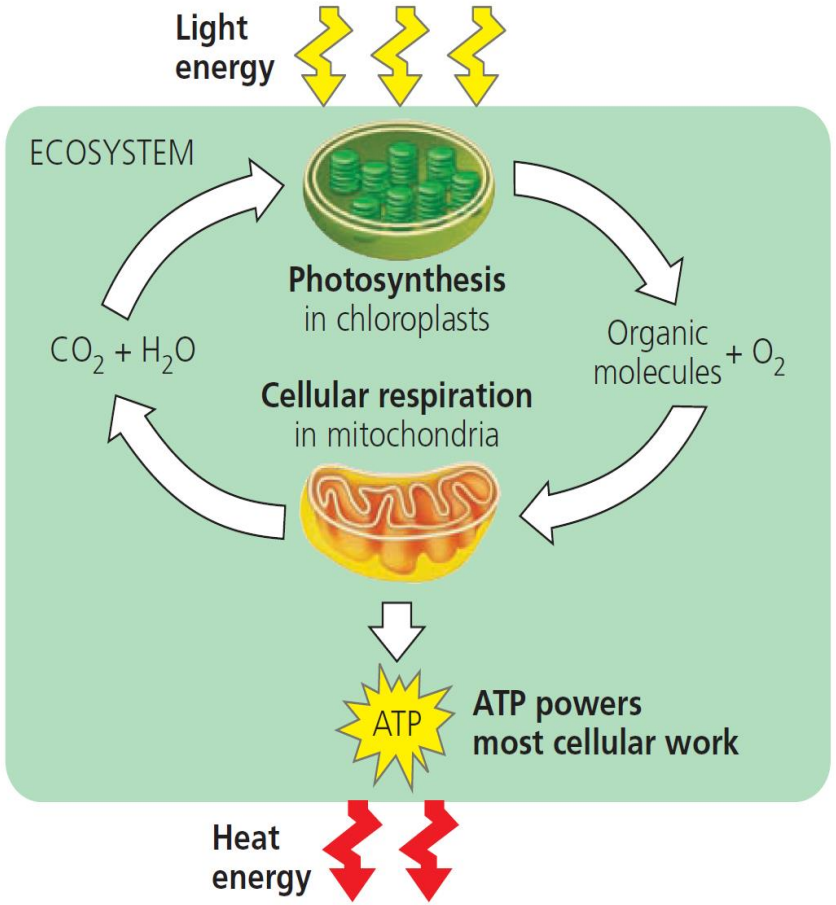
9.4 During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

9.5 Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen

9.6 Glycolysis and the citric acid cycle connect to many other metabolic pathways

Energy sources

- To perform their many tasks, **living cells require energy from outside sources**.
- Energy enters most ecosystems as sunlight and leaves as heat. In contrast, **the chemical elements essential for life are recycled**.
- **Photosynthesis** generates oxygen and organic molecules that the mitochondria of eukaryotes (including plants and algae) use as fuel for cellular respiration.
- **Cells harvest the chemical energy** stored in organic molecules and use it to regenerate ATP, the molecule that drives most cellular work.
- **Respiration has three key pathways: glycolysis, the citric acid cycle, and oxidative phosphorylation.**
- Fermentation is a simpler pathway coupled to glycolysis that has deep evolutionary roots.



▲ **Figure 9.2 Energy flow and chemical recycling in ecosystems.** Energy flows into an ecosystem as sunlight and ultimately leaves as heat, while the chemical elements essential to life are recycled.

Fermentation and aerobic respiration

- **Catabolic metabolic pathways release energy stored in complex organic molecules.**
  - ✓ *Electron transfer plays a major role in these pathways*.

- Enzymes catalyze the systematic degradation of organic molecules that are rich in energy to simpler waste products that have less energy.

- Some of the released energy is used to do work; the rest is dissipated as heat.

- One type of catabolic process, **fermentation**, leads to the *partial degradation of sugars without the use of oxygen*.

- A more efficient and widespread catabolic process, **aerobic respiration**, consumes oxygen as a reactant to complete the breakdown of a variety of organic molecules.

  - ✓ *Most eukaryotic and many prokaryotic organisms can carry out aerobic respiration.*
  - ✓ *Some prokaryotes use compounds other than oxygen as reactants in a similar process called anaerobic respiration.*
  - ✓ *Although **cellular respiration** technically includes both aerobic and anaerobic processes, the term is commonly used to refer only to the aerobic process.*

Redox reactions

- ***Redox reactions release energy when electrons move closer to electronegative atoms.***

- Catabolic pathways **transfer the electrons** stored in food molecules, releasing energy that is used to synthesize ATP.

- Reactions that result in the transfer of one or more electrons (e^-) from one reactant to another are oxidation-reduction reactions, or **redox reactions**.

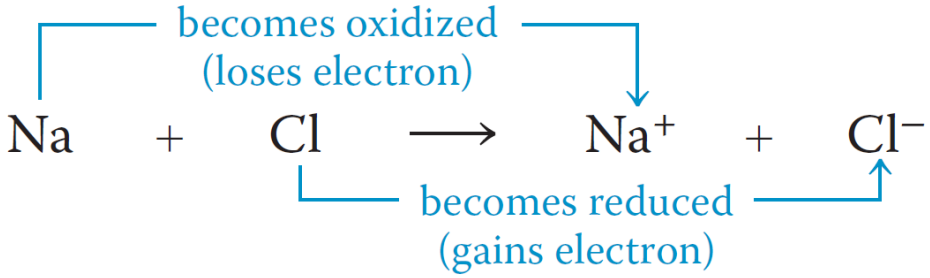
  - ✓ *The loss of electrons from a substance is called **oxidation**.*

  - ✓ *The addition of electrons to another substance is called **reduction**.*

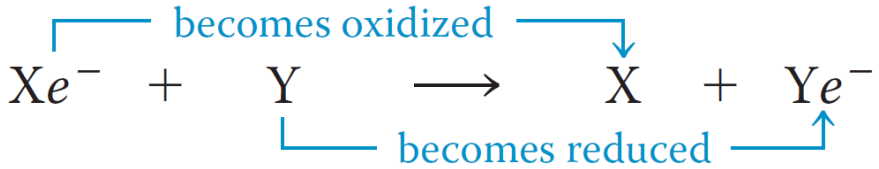
  - ✓ *Adding electrons is called reduction because negatively charged electrons added to an atom reduce the amount of positive charge of that atom.*

OIL-RIG

Redox reactions



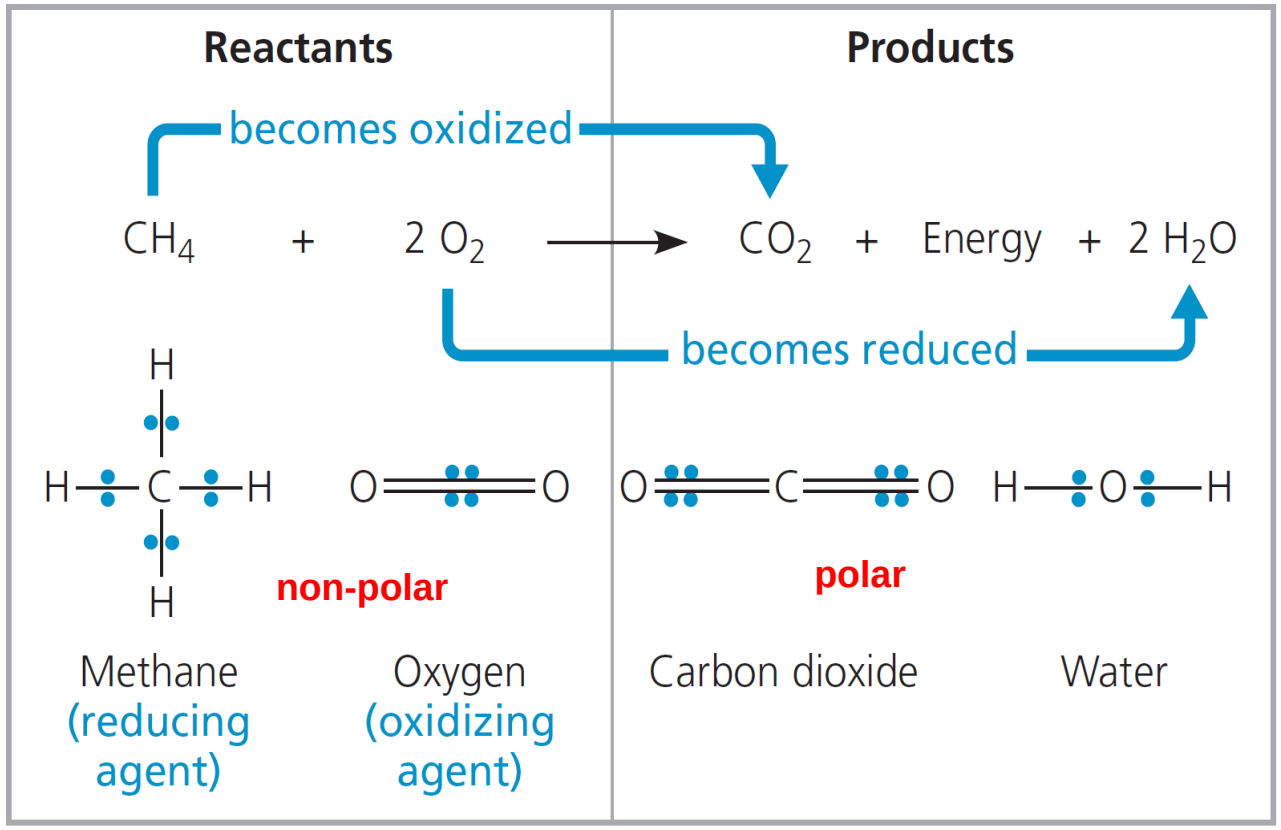
We could generalize a redox reaction this way:



❖ More generally: $Xe^- + Y \rightarrow X + Ye^-$.

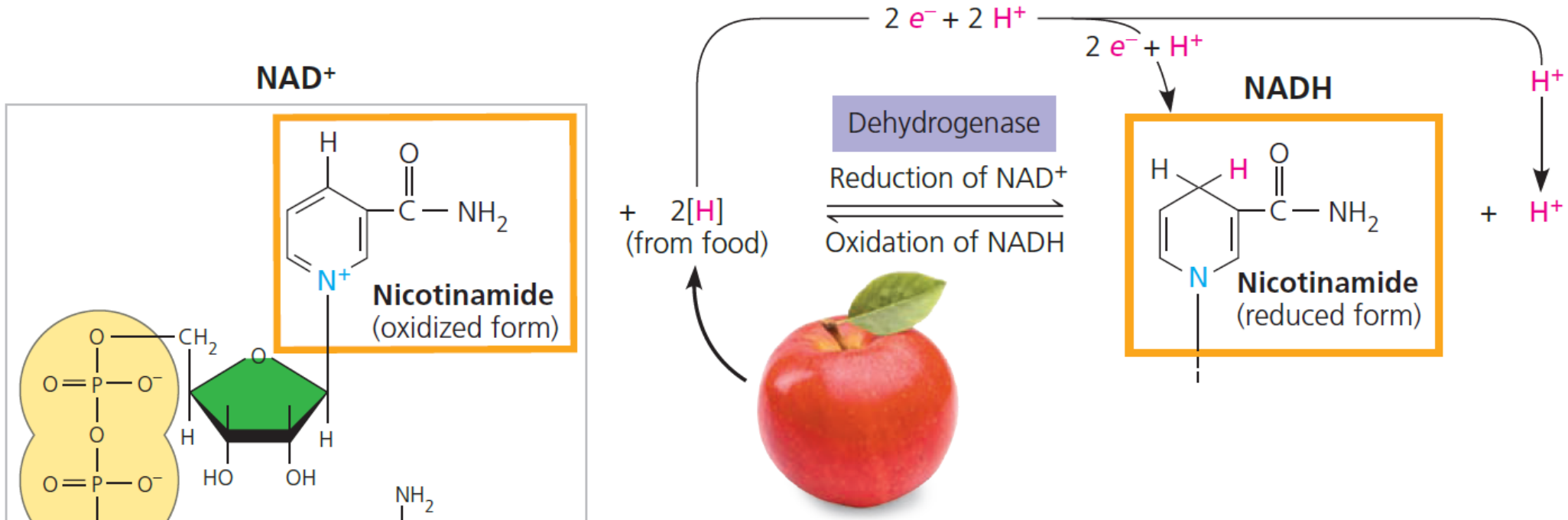
✓ *X, the electron donor, is the **reducing agent** and reduces Y by donating an electron to it.*

✓ *Y, the electron recipient, is the **oxidizing agent** and oxidizes X by removing its electron.*



▲ **Figure 9.3 Methane combustion as an energy-yielding redox reaction.** The reaction releases energy to the surroundings because the electrons lose potential energy when they end up being shared unequally, spending more time near electronegative atoms such as oxygen.

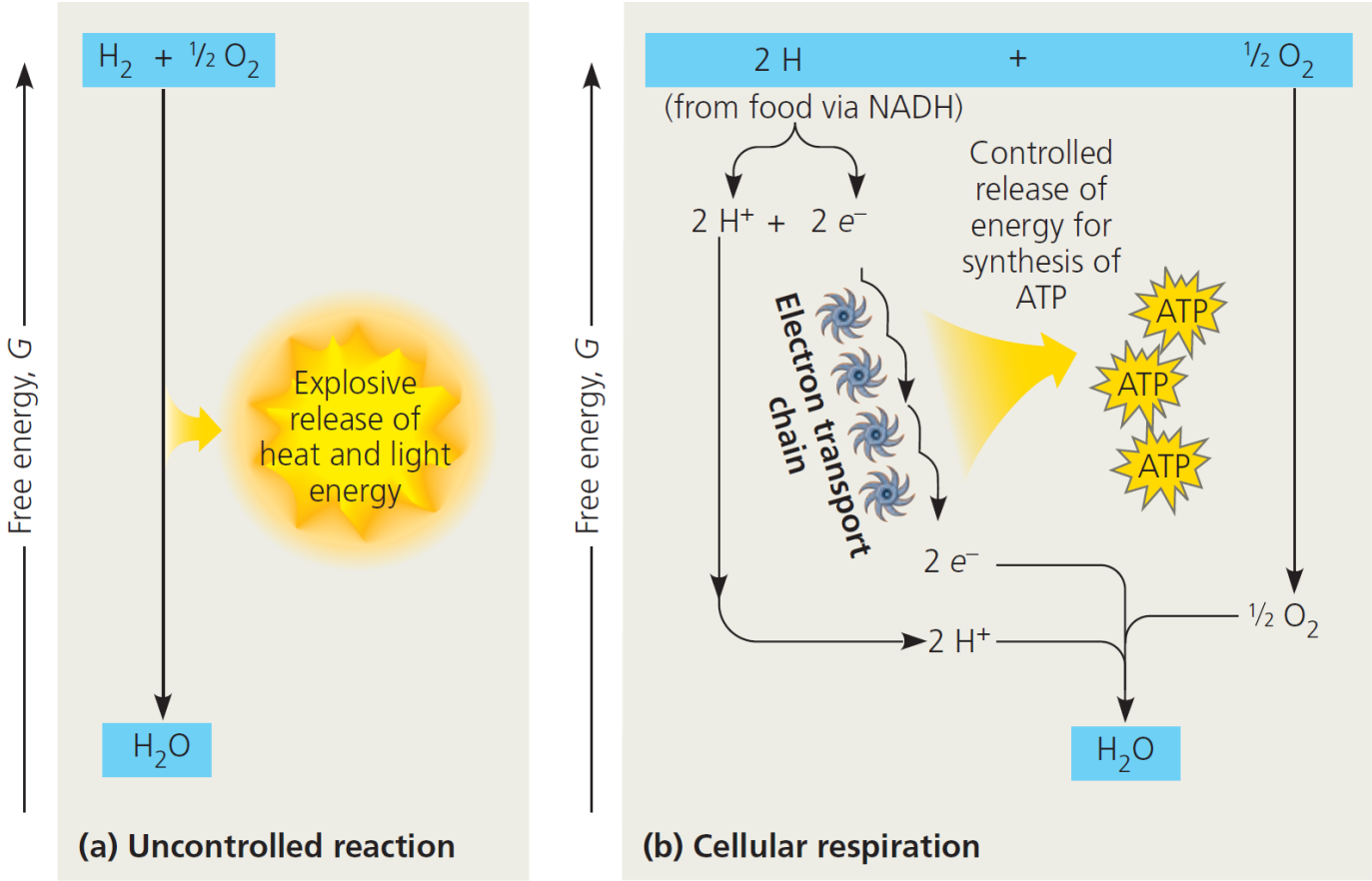
Example on the role of NAD⁺/NADH



▲ **Figure 9.4 NAD⁺ as an electron shuttle.** The full name for NAD⁺, nicotinamide adenine dinucleotide, describes its structure—the molecule consists of two nucleotides joined together at their phosphate groups (shown in yellow). (Nicotinamide is a nitrogenous base, although not one that is present in DNA or RNA; see Figure 5.24.) The enzymatic transfer of 2 electrons and 1 proton (H^+) from an organic molecule in food to NAD^+ reduces the NAD^+ to NADH: Most of the electrons removed from food are transferred initially to NAD^+, forming NADH.

? *Describe the structural differences between the oxidized form and the reduced form of nicotinamide.*

Stepwise electron transfer



▲ **Figure 9.5 An introduction to electron transport chains. (a)** The one-step exergonic reaction of hydrogen with oxygen to form water releases a large amount of energy in the form of heat and light: an explosion. **(b)** In cellular respiration, the same reaction occurs in stages: An electron transport chain breaks the "fall" of electrons in this reaction into a series of smaller steps and stores some of the released energy in a form that can be used to make ATP. (The rest of the energy is released as heat.)

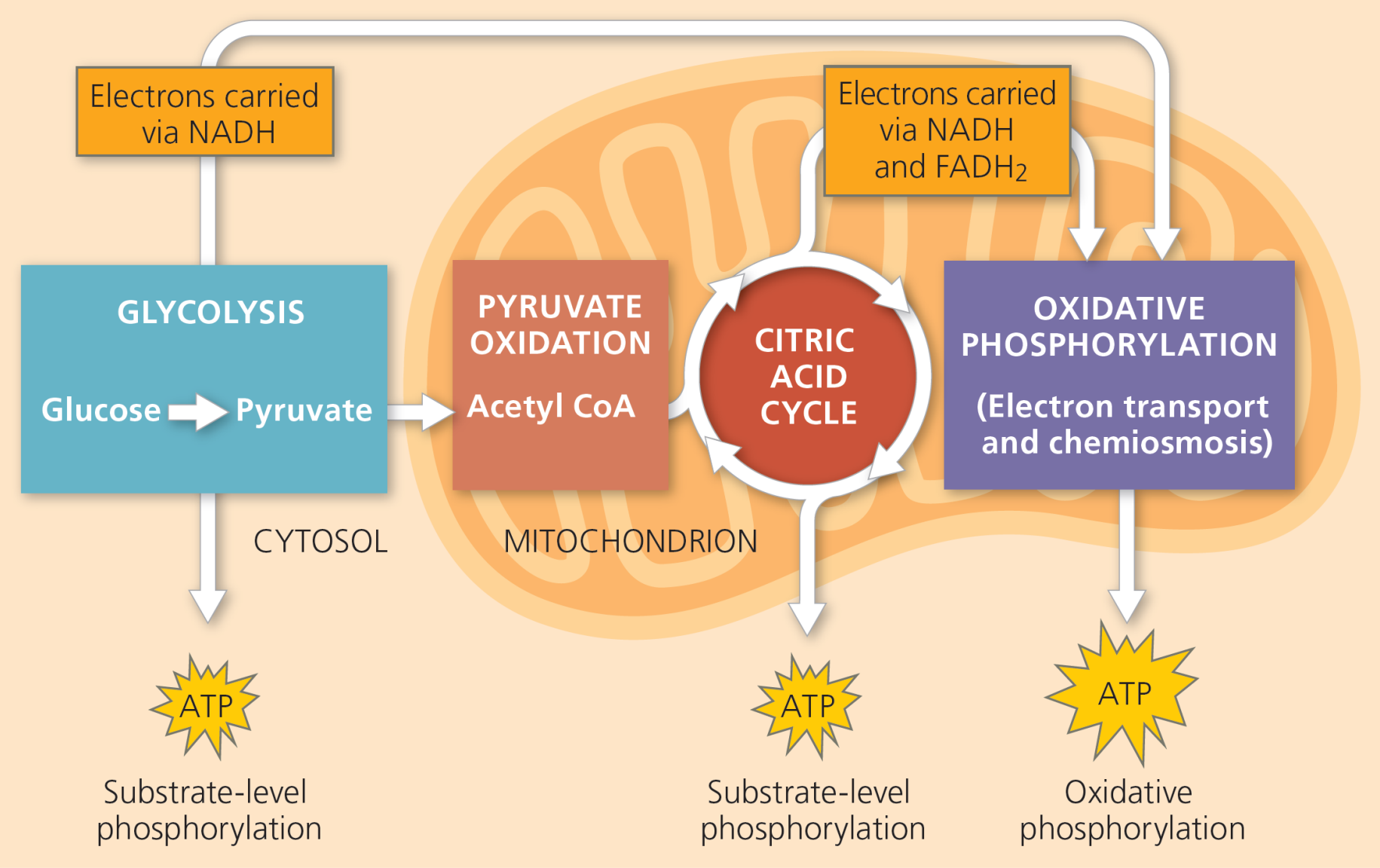
- How are electrons extracted from glucose and stored in NADH finally transferred to oxygen?
- Unlike the explosive release of heat energy that occurs when H_2 and O_2 are combined (with a spark for activation energy), cellular respiration uses an **electron transport chain** to break the fall of electrons to O_2 into several energy-releasing steps.

Catabolic pathways yield energy by oxidizing organic fuels Metabolic pathways that release stored energy

The "fall" of electrons during respiration is stepwise, via NAD^+ and an electron transport chain.

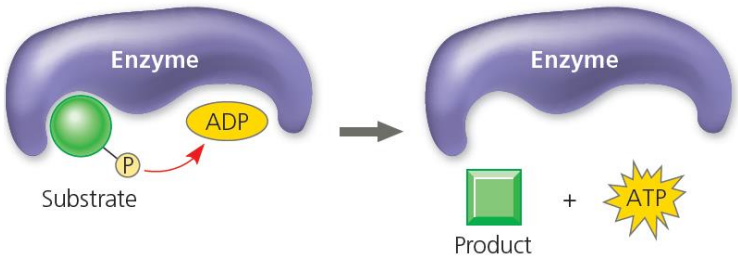
- By receiving two electrons and only one proton, NAD^+ has its charge neutralized when reduced to NADH.
  - ✓ NAD^+ functions as the oxidizing agent in many of the redox steps during the breakdown of glucose.
- The electrons carried by NADH lose very little of their potential energy in this process.
- Each NADH molecule formed during respiration represents stored energy. This energy is tapped to synthesize ATP as electrons "fall" down an energy gradient from NADH to oxygen.
- How are electrons extracted from glucose and stored in NADH finally transferred to oxygen?
- Unlike the explosive release of heat energy that occurs when H_2 and O_2 are combined (with a spark for activation energy), cellular respiration uses an **electron transport chain** to break the fall of electrons to O_2 into several energy-releasing steps.

Glycolysis: Substrate and oxidative phosphorylation



Oxidative and substrate phosphorylation

- In eukaryotic cells, the inner membrane of the mitochondrion is the site of **electron transport and chemiosmosis**, which constitute **oxidative phosphorylation**.
  - *In prokaryotes, these processes take place in the plasma membrane.*



- **Oxidative phosphorylation produces almost 90% of the ATP generated by respiration.**

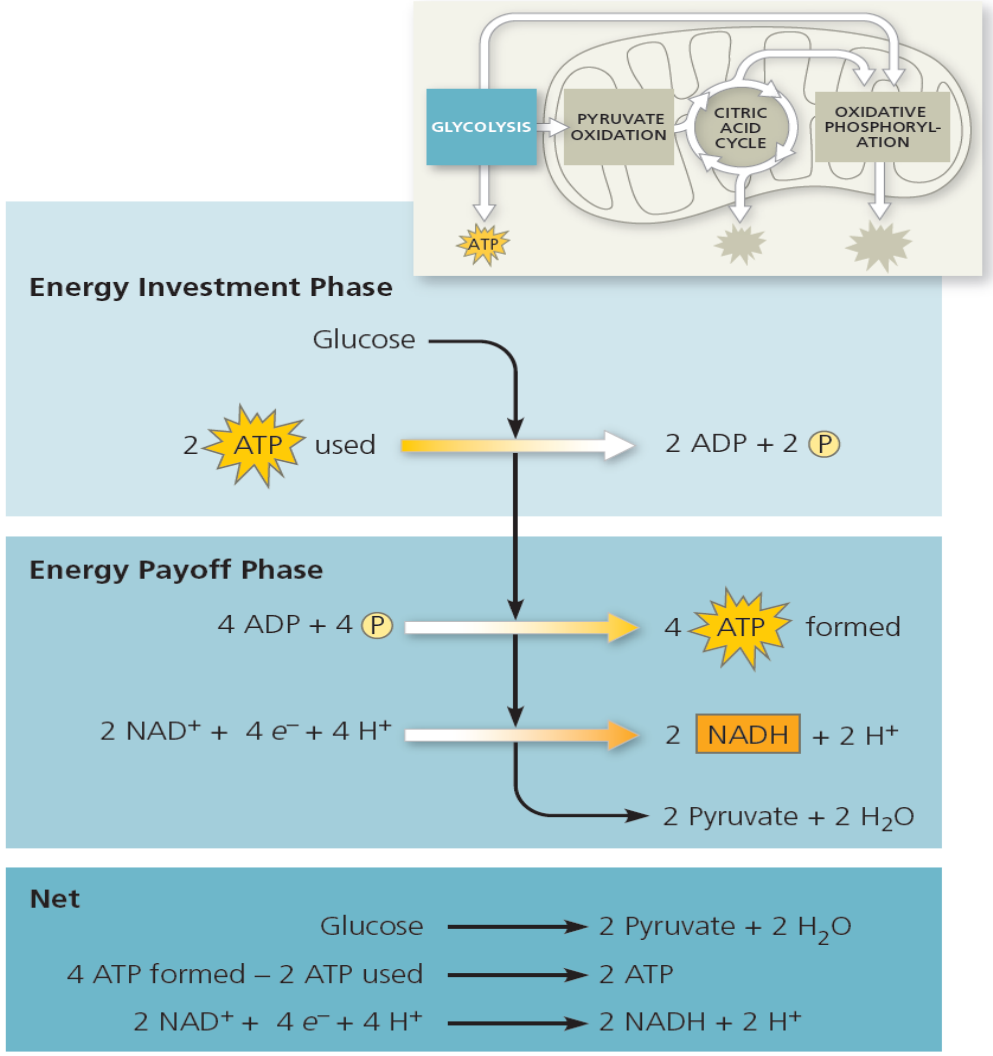
- Some ATP is also formed directly during glycolysis and the citric acid cycle by **substrate-level phosphorylation**, in which an enzyme transfers a phosphate group from an organic substrate molecule to ADP, forming ATP.

- **For each molecule of glucose degraded to carbon dioxide and water by respiration, the cell makes up to 32 ATP, each with 7.3 kcal/mol of free energy.**
- **Respiration uses the small steps in the respiratory pathway to break the large denomination of energy contained in glucose into a small change of ATP.**

  - ✓ The quantity of energy in ATP is more appropriate for the energy level of work required in the cell.

Glycolysis: 10 Steps–explanation

- During *glycolysis*, glucose, a six-carbon sugar, is split into two three-carbon sugars.
  - ✓ *These smaller sugars are then oxidized and rearranged to form two molecules of pyruvate, the ionized form of pyruvic acid.*
- Each of the ten steps in glycolysis is catalyzed by a specific enzyme.
- These steps can be divided into two phases.
  1. *In the **energy investment phase**, the cell spends ATP.*
  2. *In the **energy payoff phase**, this investment is repaid with interest. ATP is produced by substrate-level phosphorylation, and NAD^+ is reduced to NADH by electrons released by the oxidation of glucose*



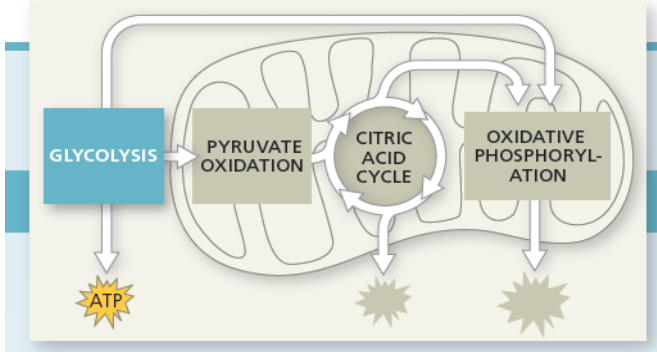
▲ **Figure 9.8** The energy input and output of glycolysis.

Explanation

- The net yield from glycolysis is 2 ATP and 2 NADH per glucose.
  - ✓ *No carbon is released as CO_2 during glycolysis.*

- Glycolysis can occur whether or not O_2 is present.
  - ✓ *If O_2 is present, the chemical energy stored in pyruvate and NADH can be extracted by pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation.*

Oxidation of Glucose to pyruvate: Steps:1-5 – ATP investment phase



▼ **Figure 9.9 A closer look at glycolysis.** Note that glycolysis is a source of ATP and NADH.

GLYCOLYSIS: Energy Investment Phase

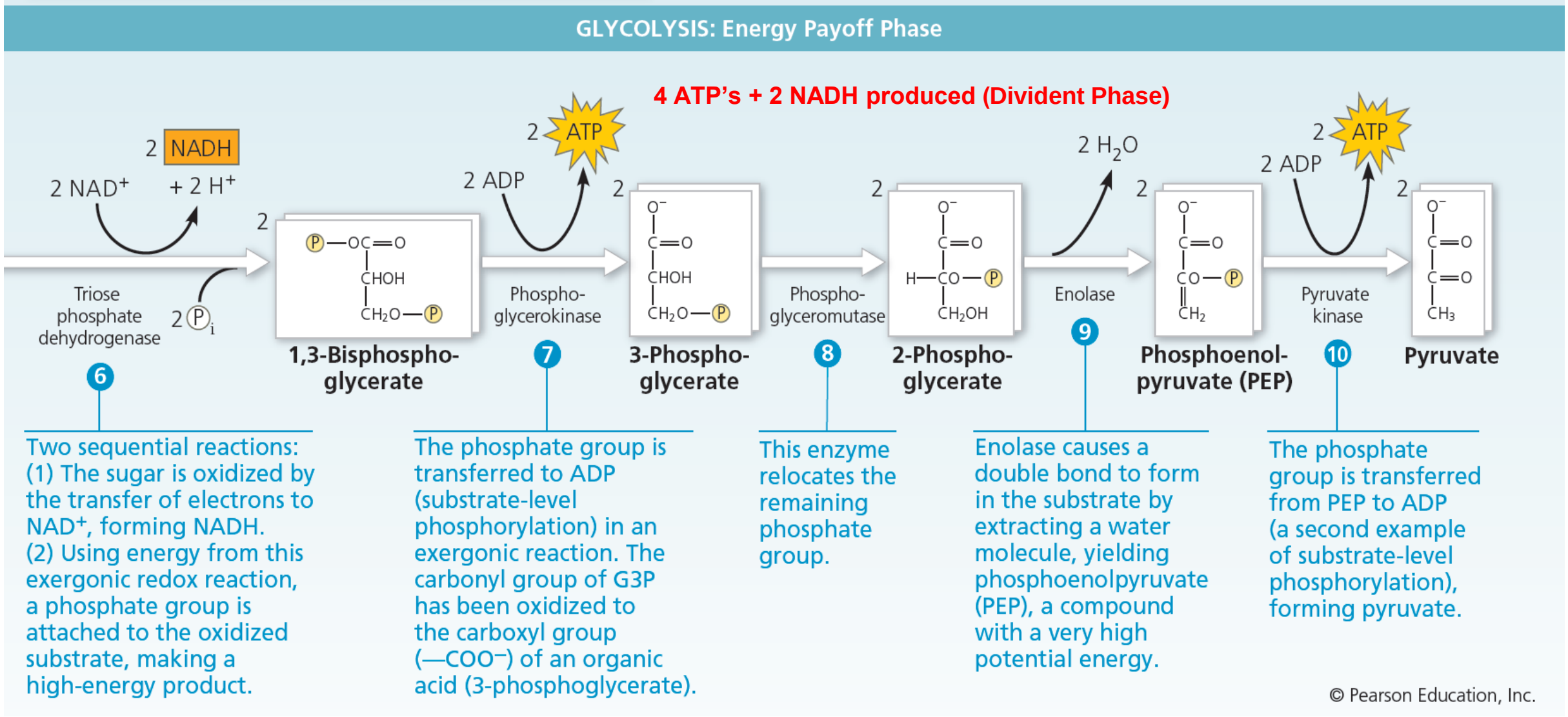
WHAT IF? *What would happen if you removed the dihydroxyacetone phosphate generated in step 4 as fast as it was produced?*

2 ATP's utilized (Investment Phase)

Glucose → ATP → ADP → **Glucose 6-phosphate** (Hexokinase, 1) → **Fructose 6-phosphate** (Phosphoglucoisomerase, 2) → ATP → ADP → **Fructose 1,6-bisphosphate** (Phosphofructokinase, 3) → (Aldolase, 4) → **Glyceraldehyde 3-phosphate (G3P)** ⇄ **Dihydroxyacetone phosphate (DHAP)** (Isomerase, 5)

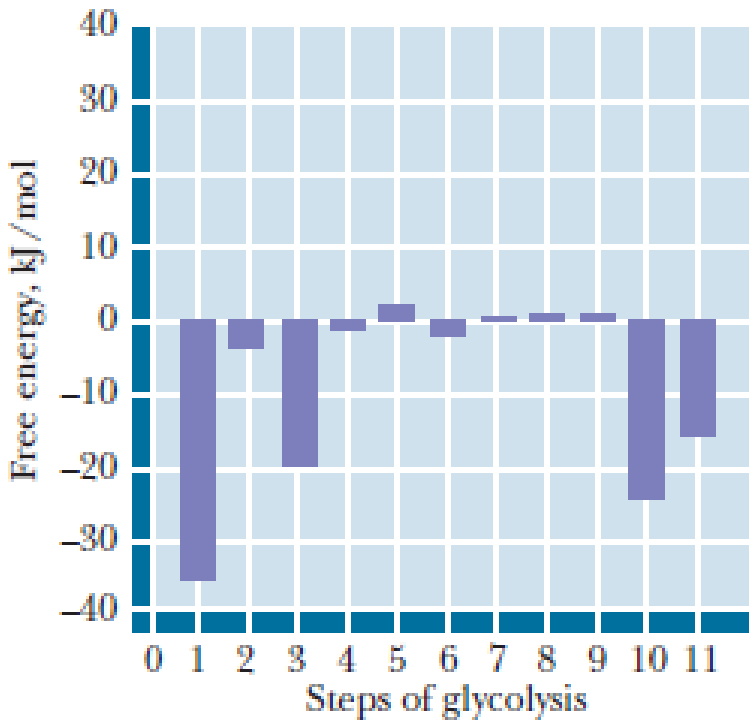
1. Hexokinase transfers a phosphate group from ATP to glucose, making it more chemically reactive. The charge on the phosphate also traps the sugar in the cell.
2. Glucose 6-phosphate is converted to fructose 6-phosphate.
3. Phosphofructokinase transfers a phosphate group from ATP to the opposite end of the sugar, investing a second molecule of ATP. This is a key step for regulation of glycolysis.
4. Aldolase cleaves the sugar molecule into two different three-carbon sugars.
5. Conversion between DHAP and G3P: This reaction never reaches equilibrium; G3P is used in the next step as fast as it forms.

Steps-6-10- ATP production



Summary of Glycolysis

Reaction	Enzyme	$\Delta G^{\circ\prime}$ (kJ/mol)	K_{eq} at 25°C	ΔG (kJ/mol)
α-D-Glucose + ATP^{4-} ⇌ glucose-6-phosphate^{2-} + ADP^{3-} + H^+	Hexokinase Glucokinase	−16.7	850	−33.9*
Glucose-6-phosphate^{2-} ⇌ fructose-6-phosphate^{2-}	Phosphoglucoisomerase	+1.67	0.51	−2.92
Fructose-6-phosphate^{2-} + ATP^{4-} ⇌ fructose-1,6-bisphosphate^{4-} + ADP^{3-} + H^+	Phosphofructokinase	−14.2	310	−18.8
Fructose-1,6-bisphosphate^{4-} ⇌ dihydroxyacetone-P^{2-} + glyceraldehyde-3-P^{2-}	Fructose bisphosphate aldolase	+23.9	6.43×10^{-5}	−0.23
Dihydroxyacetone-P^{2-} ⇌ glyceraldehyde-3-P^{2-}	Triose phosphate isomerase	+7.56	0.0472	+2.41
Glyceraldehyde-3-P^{2-} + P_i^{2-} + NAD^+ ⇌ 1,3-bisphosphoglycerate^{4-} + NADH + H^+	Glyceraldehyde-3-P dehydrogenase	+6.30	0.0786	−1.29
1,3-Bisphosphoglycerate^{4-} + ADP^{3-} ⇌ 3-P-glycerate^{3-} + ATP^{4-}	Phosphoglycerate kinase	−18.9	2060	+0.1
3-Phosphoglycerate^{3-} ⇌ 2-phosphoglycerate^{3-}	Phosphoglycerate mutase	+4.4	0.169	+0.83
2-Phosphoglycerate^{3-} ⇌ phosphoenolpyruvate^{3-} + H_2O	Enolase	+1.8	0.483	+1.1
Phosphoenolpyruvate^{3-} + ADP^{3-} + H^+ ⇌ pyruvate$^-$ + ATP^{4-}	Pyruvate kinase	−31.7	3.63×10^5	−23.0
Pyruvate$^-$ + NADH + H^+ ⇌ lactate$^-$ + NAD^+	Lactate dehydrogenase	−25.2	2.63×10^4	−14.8



- For reactions 2 and 4 through 9, Δ*G is very close to zero, meaning these reactions operate essentially at equilibrium.*
- Small changes in the concentrations of reactants and products could "push" any of these reactions either forward or backward.
- By contrast, the hexokinase, phosphofructokinase, and pyruvate kinase reactions all exhibit large negative *DG values under* cellular conditions.
- These reactions are thus the sites of glycolytic regulation. When these three enzymes are active, glycolysis proceeds and glucose is readily metabolized to pyruvate or lactate.

Summary – balance sheet

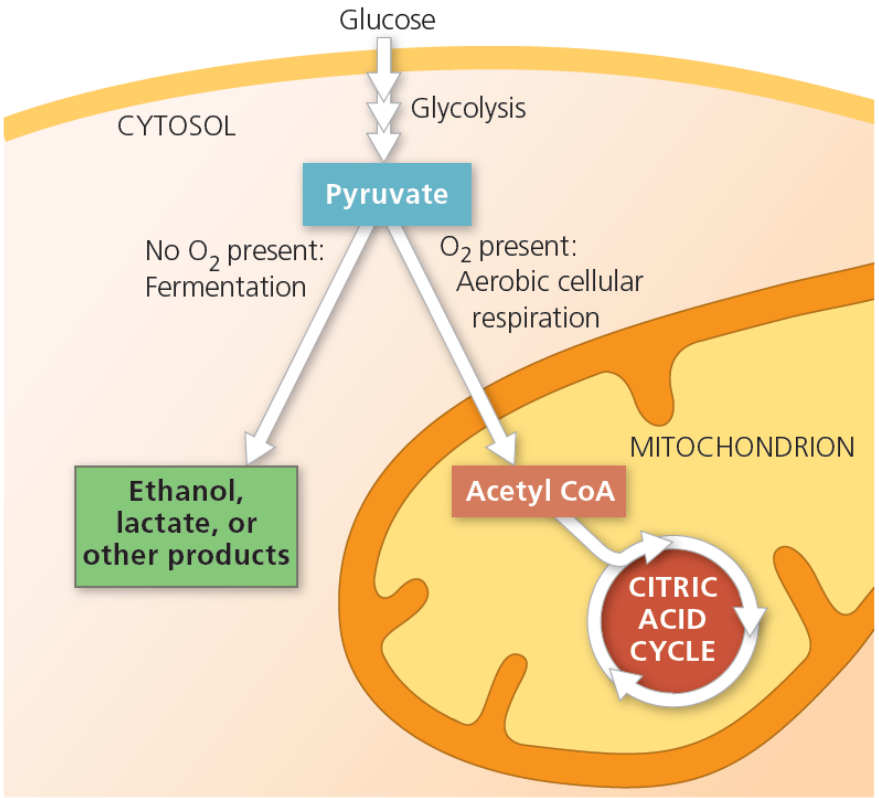
The ATP and NADH balance sheet for glycolysis

Number[a]	Reaction per Glucose	ATP Change per Glucose[b]	NADH Change per Glucose[b]
1	Glucose → glucose-6-phosphate	−1	0
3	Fructose-6-phosphate → fructose-1,6-bisphosphate	−1	0
6	2 Glyceraldehyde-3-phosphate ⇌ 2 1,3-bisphosphoglycerate	0	+2
7	2 1,3-Bisphosphoglycerate ⇌ 2 3-phosphoglycerate	+2	0
10	2 Phosphoenolpyruvate → 2 pyruvate	+2	0
Total change		+2	+2

[a] The number corresponds to the reaction number in Table 15.1.

[b] A minus sign indicates loss of ATP by cleavage of a phosphoanhydride bond; a plus sign indicates formation of ATP (from ADP) or NADH (from NAD^+).

Three fates of Pyruvate



▲ **Figure 9.18 Pyruvate as a key juncture in catabolism.** Glycolysis is common to fermentation and cellular respiration. The end product of glycolysis, pyruvate, represents a fork in the catabolic pathways of glucose oxidation. In a facultative anaerobe or a muscle cell, which are capable of both aerobic cellular respiration and fermentation, pyruvate is committed to one of those two pathways, usually depending on whether or not oxygen is present.

- Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen.

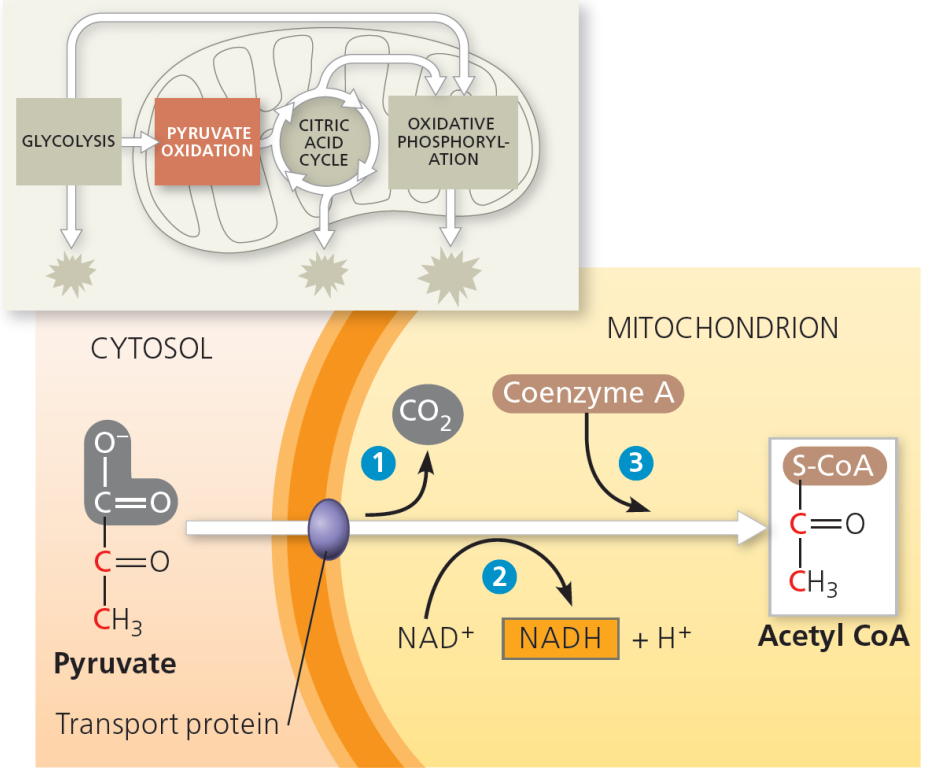
- **Human muscle cells switch from aerobic respiration to lactic acid fermentation to generate ATP when O_2 is scarce**. This may occur in the **early stages of strenuous exercise**.

  - ✓ *The waste product, lactate, was previously thought to cause muscle fatigue and pain, but recent research suggests instead that increased levels of potassium ions (K^+) may be to blame; lactate appears to enhance muscle performance.*

  - ✓ *Excess lactate is gradually carried away by the blood to the liver, where it is converted back to pyruvate by liver cells.*

  - ✓ *The pyruvate enters the mitochondria in liver cells and completes cellular respiration.*

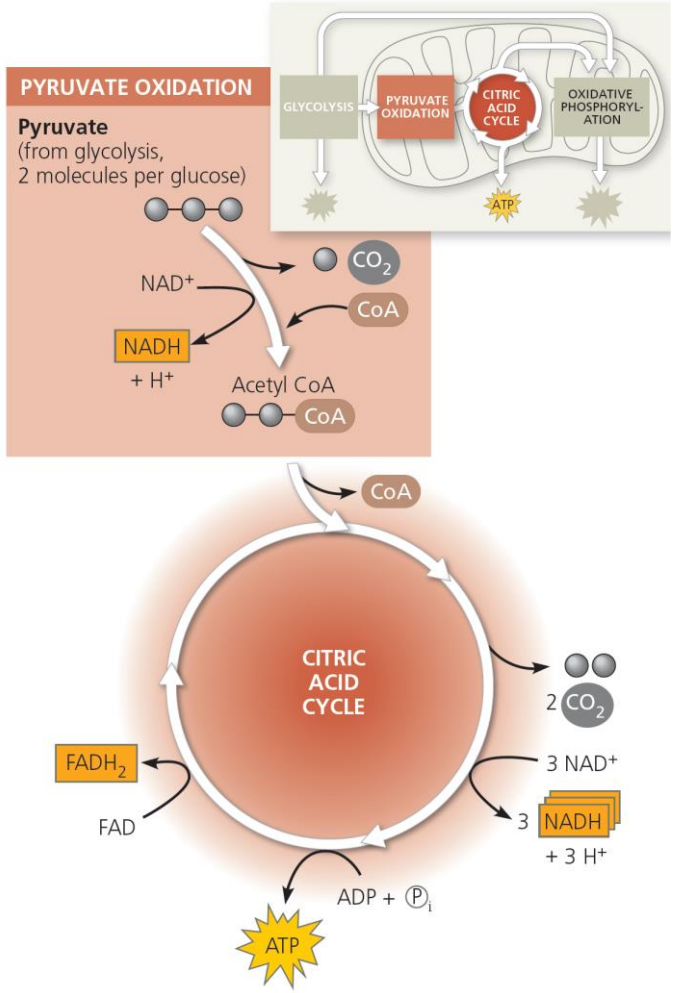
Other steps: Citric acid cycle and electron transport chain reactions



▲ **Figure 9.10 Oxidation of pyruvate to acetyl CoA, the step before the citric acid cycle.** Pyruvate is a charged molecule, so in eukaryotic cells it must enter the mitochondrion via active transport, with the help of a transport protein. Next, a complex of several enzymes (the pyruvate dehydrogenase complex) catalyzes the three numbered steps, which are described in the text. The acetyl group of acetyl CoA will enter the citric acid cycle. The CO_2 molecule will diffuse out of the cell. By convention, coenzyme A is abbreviated S-CoA when it is attached to a molecule, emphasizing the sulfur atom (S).

- ❖ Acetyl CoA now feeds its acetyl group into the citric acid cycle for further oxidation.
  - ✓ *The citric acid cycle is also called the tricarboxylic acid cycle or the **Krebs cycle**. The latter name honors Hans Krebs, who was largely responsible for elucidating the cycle's pathways in the 1930s.*
- ❖ The citric acid cycle oxidizes organic fuel derived from pyruvate.
  - ✓ *Three CO_2 molecules are released, including the one released during the conversion of pyruvate to acetyl CoA.*
  - ✓ *The cycle generates one ATP per turn by substrate-level phosphorylation.*
  - ✓ *Most of the chemical energy is transferred to NAD^+ and a related electron carrier, the coenzyme FAD, during the redox reactions.*
  - ✓ *The reduced coenzymes, NADH and $FADH_2$, transfer high-energy electrons to the electron transport chain*

Other steps: Citric acid cycle and electorn transport chain



▲ **Figure 9.11 An overview of pyruvate oxidation and the citric acid cycle.** The inputs and outputs per pyruvate molecule are shown. To calculate on a per-glucose basis, multiply by 2, because each glucose molecule is split during glycolysis into two pyruvate molecules.

- The citric acid cycle has eight steps, each catalyzed by a specific enzyme.
- The acetyl group of acetyl CoA joins the cycle by combining with the compound oxaloacetate, forming citrate.
  - ✓ *The next seven steps decompose the citrate back to oxaloacetate.*
  - ✓ *It is the regeneration of oxaloacetate that makes this process a cycle.*

Pyruvate oxidation: Citric acid cycle–oxidation to produce energy yielding cofactors

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Pyruvate oxidation: Citric acid cycle–oxidation to produce energy yielding cofactors

- For each acetyl group that enters the cycle, 3 NAD^+ are reduced to NADH.

- In one step, electrons are transferred to FAD instead of NAD^+. FAD then accepts 2 electrons and 2 protons to become $FADH_2$.

- In the cells of plants, bacteria, and some animal tissues, the citric acid cycle forms an ATP molecule by substrate-level phosphorylation.

- In many animal tissue cells, guanosine triphosphate (GTP) is formed by the same process of substrate-level phosphorylation.

  - ✓ GTP may be used to synthesize ATP or directly power work in the cell.
  - ✓ The output from this step is the only ATP generated directly by the citric acid cycle.
  - ✓ The total yield per glucose from the citric acid cycle is 6 NADHs, 2 FADHs, and the equivalent of 2 ATPs.

Continued

- **Most of the ATP produced by respiration results from oxidative phosphorylation, when the NADH and $FADH_2$ produced by the citric acid cycle relay the electrons extracted from food to the electron transport chain.**
  - ✓ This process supplies the necessary energy for the phosphorylation of ADP to ATP.
- Only 4 of 38 ATP produced by the respiration of glucose are produced by substrate-level phosphorylation: 2 net ATP from glycolysis and 2 ATP from the citric acid cycle.
- NADH and $FADH_2$ account for most of the energy extracted from glucose.
  - ✓ These reduced coenzymes link glycolysis and the citric acid cycle to oxidative phosphorylation, which uses energy released by the electron transport chain to power ATP synthesis.

Summary of citric acid cycle

Reactions of the TCA Cycle

Reaction	Enzyme	$\Delta G^{\circ\prime}$ (kcal/mol)
1. Acetyl-CoA + oxaloacetate + H_2O $\longrightarrow$ citrate + CoA	Citrate synthase	−7.7
2. Citrate $\rightleftharpoons$ *cis*-aconitase $\rightleftharpoons$ isocitrate	Aconitase	+1.5
3. Isocitrate + NAD^+ $\longrightarrow$ α-ketoglutarate + NADH + CO_2	Isocitrate dehydrogenase	−5.0
4. α-Ketoglutarate + NAD^+ + CoA $\longrightarrow$ Succinyl-CoA + NADH + CO_2	α-Ketoglutarate dehydrogenase	−8.0
5. Succinyl-CoA + GDP + P_i $\longrightarrow$ succinate + GTP + CoA	Succinate thiokinase	−0.7
6. Succinate + FAD $\longrightarrow$ fumarate + $FADH_2$	Succinate dehydrogenase	0.0
7. Fumarate + H_2O $\longrightarrow$ L-malate	Fumarase	−0.9
8. L-malate + NAD^+ $\longrightarrow$ oxaloacetate + NADH + H^+	Malate dehydrogenase	+7.1
Acetyl-CoA + 2 H_2O + 3 NAD^+ + FAD + ADP + P_i $\longrightarrow$ 2 CO_2 + 3 NADH + 2 H^+ + $FADH_2$ + CoA + ATP		−13.7

2* (3 NADH + $FADH_2$ + 1 ATP)

Oxidative phosphorylation: Coupling of chemiosmosis and electron transport

- ***The inner mitochondrial membrane couples electron transport to ATP synthesis.***

- **The electron transport chain is a collection of molecules embedded in the cristae, the folded inner membrane of the mitochondrion.**
  - ✓ In prokaryotes, the electron transport chain is located in the plasma membrane.

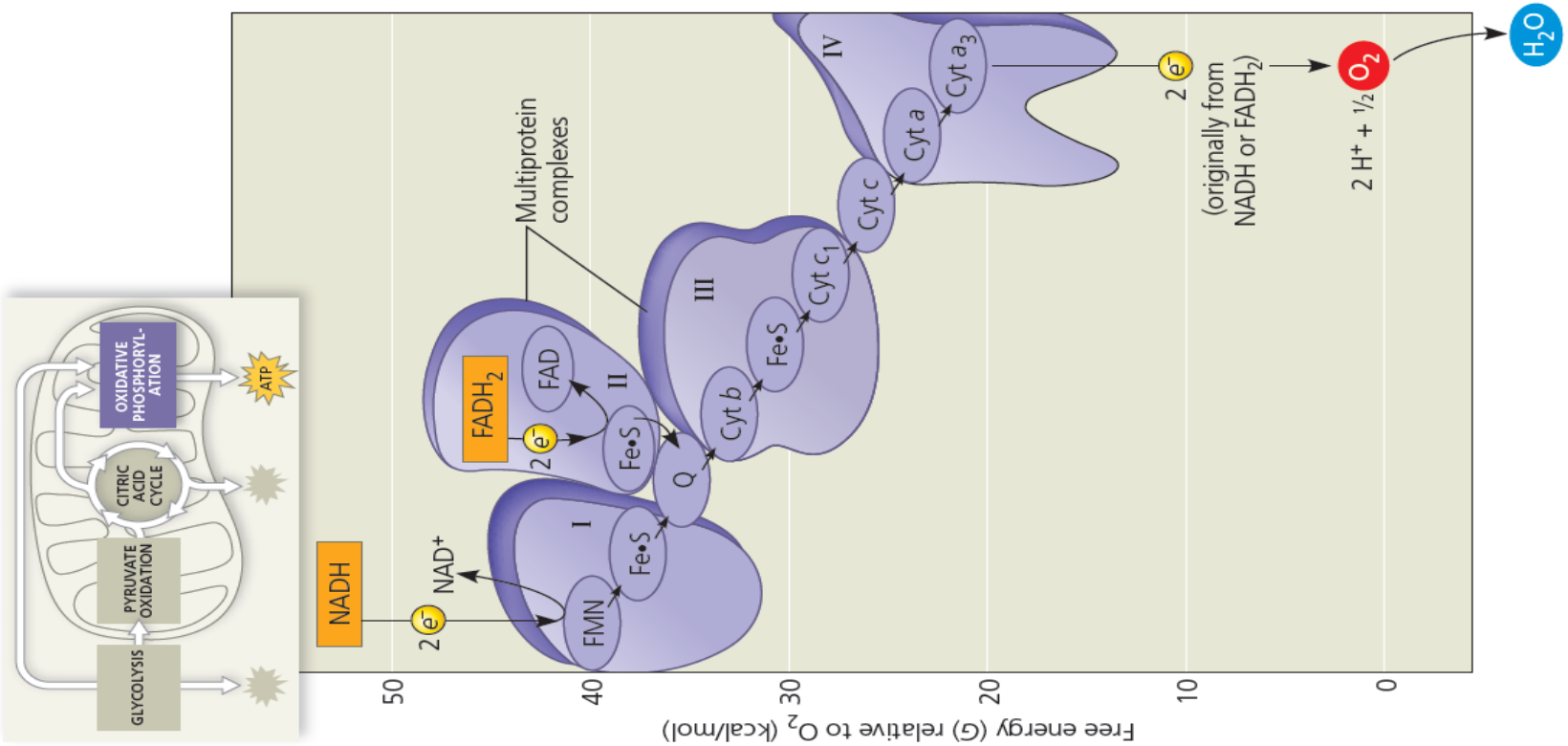
- The folding of the inner membrane to form cristae increases its surface area, providing space for thousands of copies of the chain in each mitochondrion.

- Most components of the chain are proteins that exist in multiprotein complexes numbered I– IV.
  - ✓ Tightly bound to these proteins are *prosthetic groups,* nonprotein components essential for catalysis.

- Electrons drop in free energy as they pass down the electron transport chain.

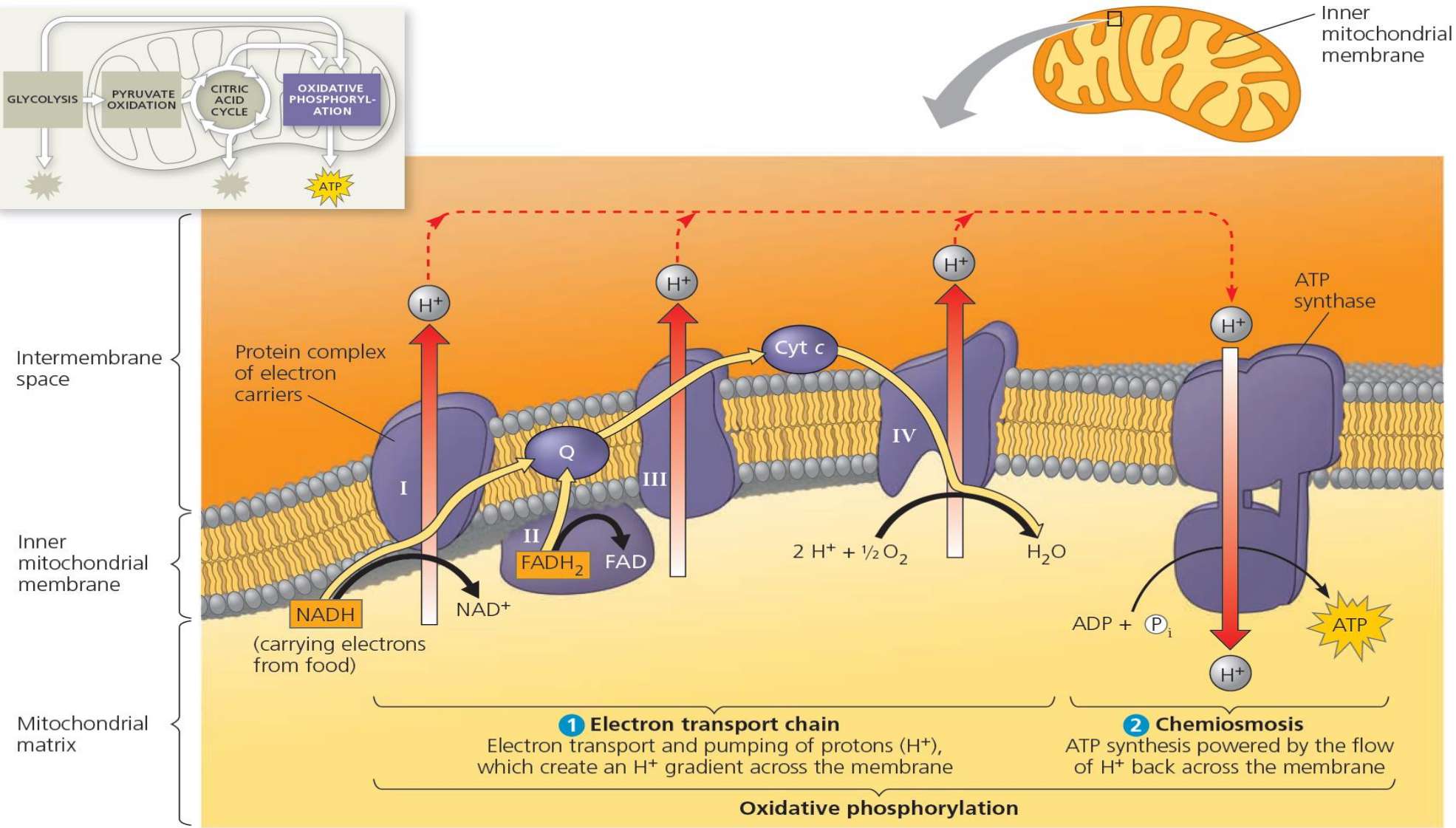
Electron transport chain (etc)

- During electron transport along the chain, electron carriers alternate between reduced and oxidized states as they accept and donate electrons.
  - ✓ Each component of the chain becomes reduced when it accepts electrons from its "uphill" neighbor, which is less electronegative.
  - ✓ It then returns to its oxidized form as it passes electrons to its more electronegative "downhill" neighbor.



▲ **Figure 9.13 Free-energy change during electron transport.** The overall energy drop (ΔG) for electrons traveling from NADH to oxygen is 53 kcal/mol, but this "fall" is broken up into a series of smaller steps by the electron transport chain. (An oxygen atom is represented here as ½ O_2 to emphasize that the electron transport chain reduces molecular oxygen, O_2, not individual oxygen atoms.)

Electron transport chain



Movement of electrons through complexes

- Electrons carried by NADH are transferred to the first molecule in the electron transport chain, a flavoprotein.

- In the next redox reaction, the flavoprotein returns to its oxidized form as it passes electrons to an iron-sulfur protein.

- The iron-sulfur protein then passes the electrons to a compound called ubiquinone, a small hydrophobic molecule and the only member of the electron transport chain that is not a protein.
  - ✓ Ubiquinone is individually mobile within the membrane rather than residing in a particular complex.

- Most of the remaining electron carriers between ubiquinone and oxygen are proteins called **cytochromes.**
  - ✓ The prosthetic group of each cytochrome is a heme group with an iron atom that accepts and donates electrons.

- The last cytochrome of the chain, cyt a_3, passes its electrons to oxygen, which is *very* electronegative.
  - ✓ Each oxygen atom also picks up a pair of hydrogen ions from the aqueous solution to form water.

Continued….

- The electrons carried by $FADH_2$ have lower free energy and are added at a lower energy level than those carried by NADH.
  - The electron transport chain provides about one-third less energy for ATP synthesis when the electron donor is $FADH_2$ rather than NADH.
- The electron transport chain generates no ATP directly.
- Its function is to break the large free-energy drop from food to oxygen into a series of smaller steps that release energy in manageable amounts.

ATP synthase: Important complex

Chemiosmosis couples electron transport and energy release to ATP synthesis.

- A protein complex in the cristae, **ATP synthase**, actually makes ATP from ADP and inorganic phosphate.

- **ATP synthase works like an ion pump running in reverse.**

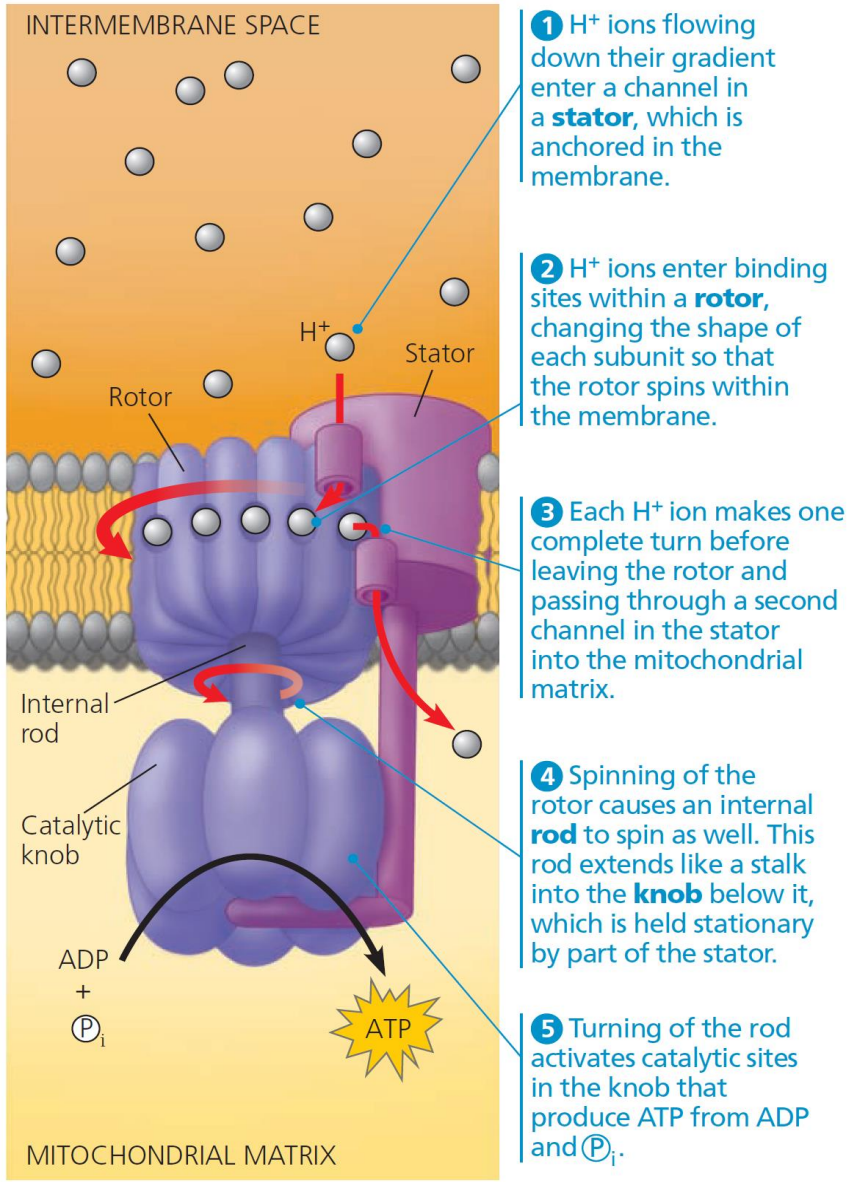
  - Ion pumps usually use ATP as an energy source to transport ions against their gradients.

- Enzymes can catalyze a reaction in either direction, depending on the ΔG for the reaction, which is affected by the local concentrations of reactants and products.

Chemiosmosis

- Rather than hydrolyzing ATP to pump protons against their concentration gradient, under the conditions of cellular respiration, ATP synthase uses the energy of an existing ion gradient to power ATP synthesis.
  - ✓ The power source for the ATP synthase is a difference in the concentrations of H^+ on opposite sides of the inner mitochondrial membrane. This is also a pH gradient.
- This process, in which *energy stored in the form of a hydrogen ion gradient across a membrane is used to drive cellular work such as the synthesis of ATP*, is called **chemiosmosis**.
  - ✓ Here, *osmosis* refers to the flow of H^+ across a membrane.

ATP synthase and molecular motor



- From studying **<u>the structure of ATP synthase</u>**, scientists have learned how the flow of H^+ through this large enzyme powers ATP generation.
- **ATP synthase is a multisubunit complex with four main parts, each made up of multiple polypeptides.**
  - ✓ Protons move one by one into binding sites on one of the parts (the rotor), causing it to spin in a way that catalyzes ATP production from ADP and inorganic phosphate.
  - ✓ ATP synthase is the smallest molecular rotary motor known in nature.
- How does the inner mitochondrial membrane or the prokaryotic plasma membrane generate and maintain the H^+ gradient that drives ATP synthesis in the ATP synthase protein complex?
  - ✓ Establishing the H^+ gradient is the function of the electron transport chain.
  - ✓ The chain is an energy converter that uses the exergonic flow of electrons to pump H^+ across the membrane from the mitochondrial matrix into the intermembrane space.
  - ✓ The H^+ has a tendency to diffuse down its gradient.

Final balance sheet– No of ATP's from 1 mole of glucose

Balance Sheet for Oxidation of One Molecule of Glucose

	NADH	$FADH_2$	ATP
CYTOPLASMIC REACTIONS			
Glycolysis			
1. Glucose ⟶ glucose-6-P			−1
2. Fructose-6-P ⟶ fructose-1,6-P			−1
3. (2) Glyceraldehyde-3-P ⟶ glycerate-1,3-bisphosphate	+2		
4. (2) Glycerate-1,3-bisphosphate ⟶ glycerate-3-P			+2
5. (2) Phosphoenolpyruvate ⟶ pyruvate			+2
MITOCHONDRIAL REACTIONS			
Pyruvate Dehydrogenase			
(2) Pyruvate ⟶ acetyl-CoA	+2		
TCA Cycle			
1. (2) Isocitrate ⟶ α-ketoglutarate	+2		
2. (2) α-Ketoglutarate ⟶ succinyl-CoA	+2		
3. (2) Succinyl-CoA ⟶ succinate			+1.5[a]
4. (2) Succinate ⟶ fumarate		+2	
5. (2) Malate ⟶ oxaloacetate	+2		
Electron Transport–Oxidative Phosphorylation			
1. Reoxidation of NADH produced in glycolysis	−2		+3[b]
2. Reoxidation of NADH produced by pyruvate dehydrogenase	−2		+5
3. Reoxidation of NADH produced in TCA cycle	−6		+15
4. Reoxidation of $FADH_2$ produced in TCA cycle		−2	+3
Totals	0	0	29.5

[a] This number is 1.5 instead of 2 to take into account the energy required to export ATP to the cytosol.

[b] This number is 3 instead of 5 because the glycerol phosphate shuttle bypasses complex I.

Calculation of number of ATPs.

1. Glycolysis = 2 ATP + 2 NADH
   = 2 ATP + 6 ATP = 8 ATPs

Note:

1 NADH = 3 ATP.
1 FADH2 = 2 ATP

2. TCA + ETC cycle
   = 6 NADH + 2 FADH2 + 2 ATP
   = 6(3 ATP) + 2 (2 ATP) + 2 ATP
   = 18 + 4 + 2 ATPs
   = 26 ATPs

Total ATP is 8 + 26 = 34 ATPs

How many ATP's are produced from Kreb's cycle (one cycle)

- The number of ATP molecules produced by the electron transport system from Krebs cycle intermediates for once cycle can be calculated.

- A cycle for Krebs cycle produces 3 molecules of reduced coenzyme NADH2 and 1 molecule of reduced FADH2 besides 1 molecule of GTP is synthesized directly through substrate level phosphorylation (3:1:1).

- The reduced coenzymes (NADH2 and FADH2) are oxidised by mitochondrial electron transport system in the inner mitochondrial membrane.

✓ *A molecule of NADH2 when undergoes oxidation produces 3 molecules of ATP*
✓ *A molecule of FADH2 when undergoes oxidation produces 2 molecules of ATP*

- Thus the total ATP produced on oxidation of reduced coenzymes produced in one turn of Krebs cycle are (3x3) + (2x1) + 1 i.e., 12 ATP molecules.

Why ATP number varies?

The number of ATP molecules generated from the catabolism of glucose varies. The reasons are

1. The number of hydrogen ions that the electron transport chain complexes can pump through the membrane varies between species.

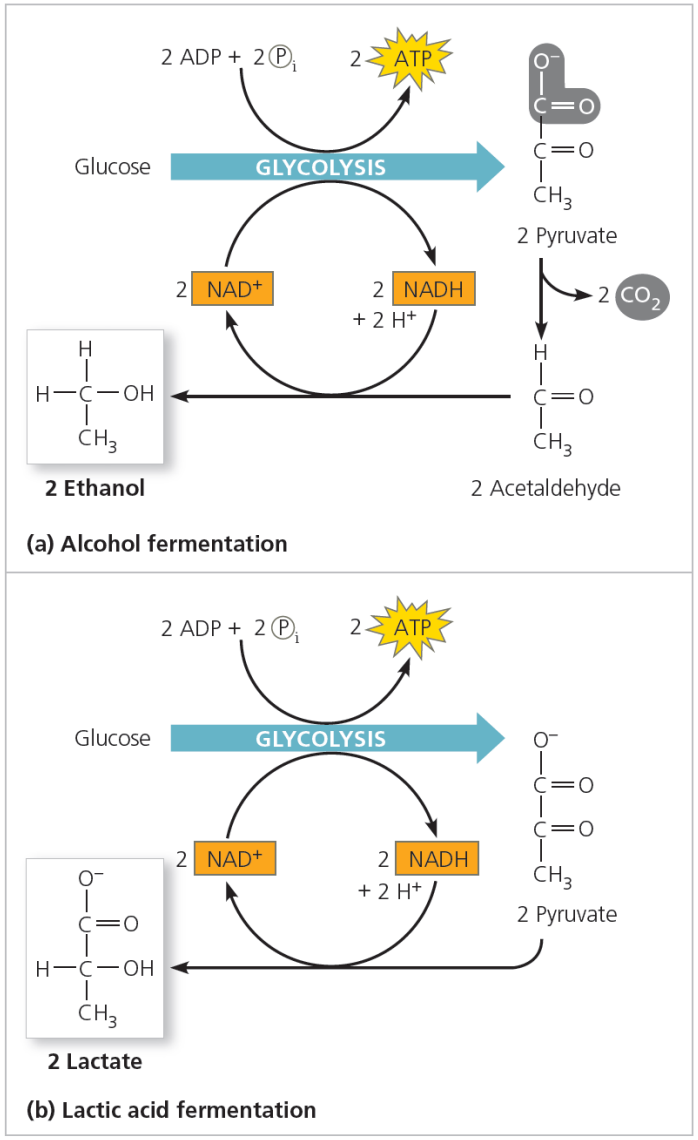
2. Another source of variance stems from the shuttle of electrons across the mitochondrial membrane. The NADH generated from glycolysis cannot easily enter mitochondria. Thus, electrons are picked up on the inside of the mitochondria by either NAD^+ or FAD^+.

3. Fewer ATP molecules are generated when FAD^+ acts as a carrier. NAD^+ is used as the electron transporter in the liver and FAD^+ in the brain, so ATP yield depends on the tissue being considered.

4. Another factor that affects the yield of ATP molecules generated from glucose is that intermediate compounds in these pathways are used for other purposes. Glucose catabolism connects with the pathways that build or break down all other biochemical compounds in cells, and the result is somewhat messier than the ideal situations described thus far.

5. For example, sugars other than glucose are fed into the glycolytic pathway for energy extraction. Other molecules that would otherwise be used to harvest energy in glycolysis or the citric acid cycle may be removed to form nucleic acids, amino acids, lipids, or other compounds. Overall, in living systems, these pathways of glucose catabolism extract about 34 percent of the energy contained in glucose.

Fermentation– Fate of Pyruvate



Fermentation pathways recycle NAD^+ by transferring electrons from NADH to pyruvate or derivatives of pyruvate.

- In **alcohol fermentation**, pyruvate is converted to ethanol in two steps.
  - ✓ Pyruvate is converted to a two-carbon compound, acetaldehyde, by the removal of CO_2.
  - ✓ Acetaldehyde is reduced by NADH to **ethanol**. This process regenerates the supply of NAD^+ needed for the continuation of glycolysis.
  - ✓ **Alcohol fermentation by yeast is used in brewing, baking, and winemaking**.

- During **lactic acid fermentation**, pyruvate is reduced directly by NADH to form lactate (the ionized form of lactic acid) without the release of CO_2.
  - ✓ Lactic acid fermentation by some fungi and bacteria is used to make cheese and yogurt.

Cellular pathways to produce ATP: ***Fermentation and cellular respiration are compared.***

- **Fermentation, anaerobic respiration, and aerobic respiration are three alternative cellular pathways for producing ATP from sugars.**

  - ✓ *All three use glycolysis to oxidize sugars to pyruvate with a net production of 2 ATP by substrate-level phosphorylation.*
  - ✓ *In all three, NAD^+ is the oxidizing agent that accepts electrons from food during glycolysis.*

- A key difference is *the mechanisms for oxidizing NADH to NAD^+,* which is required to sustain glycolysis.

  - ✓ *In fermentation, the final electron acceptor is an organic molecule such as pyruvate (lactic acid fermentation) or acetaldehyde (alcohol fermentation).*
  - ✓ *In cellular respiration, electrons carried by NADH are transferred to an electron transport chain and move stepwise down a series of redox reactions to a final electron acceptor.*

- In aerobic respiration, the *final electron acceptor* is oxygen; in anaerobic respiration, the final acceptor is another molecule that is less electronegative than oxygen.

Continued…

- Transfer of electrons from NADH to the electron transport chain not only regenerates the NAD^+ required for glycolysis but also pays an ATP bonus when the stepwise electron transport from this NADH to oxygen drives oxidative phosphorylation.

- More ATP is produced by the oxidation of pyruvate in the mitochondrion, which is unique to respiration.
  - ✓ *Without an electron transport chain, the energy still stored in pyruvate is unavailable to most cells.*
  - ✓ *Thus,* ***cellular respiration harvests much more energy from each sugar molecule than fermentation can****.*

- **Aerobic respiration yields up to 16 times as much ATP per glucose molecule as does fermentation—up to 32 molecules of ATP for respiration, compared with 2 molecules of ATP produced by substrate-level phosphorylation in fermentation.**